



DOUGLAS COLLEGE

CSIS 4495-002 (Applied Research Project)

Progress Report 4

Project name: **EcoTots**

Submitted To:

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Work Log Table

Date	Number of Hours	Description of Work done
March 25	2	Searched, chose and started Ollama for the AI integration.
March 28	3	Completed the implementation of API and backend working of normal chatbot.
March 29	2	Connected chatbot to frontend and made UI at homepage.
March 30	3	Worked on Model, Controller, Route for MongoDB.
March 30	1	Worked on Progress Report 4.

Description of work done:

I began by researching and selecting **Ollama** for AI integration, evaluating its capabilities and suitability for our chatbot implementation. Once finalized, I initiated the setup process to integrate Ollama into our project.

Next, I focused on developing the backend for the chatbot, implementing the **API and backend logic** to enable basic chatbot functionality. This step involved ensuring smooth communication between the chatbot and the backend services.

Following the backend development, I worked on the **frontend integration**, successfully connecting the chatbot to the UI. Additionally, I designed and refined the **homepage UI**, ensuring an intuitive and user-friendly interface.

Further progress was made on the chatbot's **Model, Controller, and Route**, structuring the chatbot's logic to connect to MongoDB (which is not completed yet) and request handling for efficient interactions. This step helped streamline the chatbot's workflow and improved response accuracy.

Finally, I documented the latest updates and completed **Progress Report 4**, summarizing key developments, completed tasks, and next steps to maintain a clear record of progress.

I faced few problems while doing my tasks, they are as follows:

AI Integration Issues – Faced difficulties in setting up **Ollama**, requiring extra time to understand its configuration and compatibility.

Backend Communication – Encountered issues with API request handling and response formatting, needing debugging and refinements.

Frontend Integration – Faced challenges with **state management and UI updates**, requiring adjustments for proper chatbot responses.

Routing & Controller Complexity – Organizing the **Model, Controller, and Routes** efficiently was tricky, requiring logic refinements.

Repo Check-in

The files/folders I have checked into the repo are as follows:

- Integrating Ollama AI:
 - Implementation/ecoTots/api/controllers/chatbot.controller.js
 - Implementation/ecoTots/api/index.js
 - Implementation/ecoTots/api/routes/chatbot.route.js
 - Implementation/ecoTots/client/src/App.jsx
 - Implementation/ecoTots/client/src/pages/ChatbotPage.jsx
- Error Handling and Moving Chatbot to Homepage:
 - Implementation/ecoTots/client/src/pages/Home.jsx
 - Implementation/ecoTots/client/src/pages/ChatbotPage.jsx
- Making Ollama Connection with MongoDB (**NOT COMPLETED YET**):
 - Implementation/ecoTots/api/controllers/chatbot.controller.js
 - Implementation/ecoTots/api/models/chatbot.model.js
 - Implementation/ecoTots/api/routes/chatbot.route.js